The empirical recurrence for OEIS A251227, recovered from its tail

Adrian Perez Fontelles
Independent researcher

8 September 2026

Abstract

OEIS A251227 records “Empirical recurrence of order 99” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 256$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 102$. Its last coefficient is $c_{99} = 30462644268000000 \neq 0$, so no further tail satisfies anything shorter; any order-99 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A251227 is “Number of (n+1)X(7+1) 0..1 arrays with no 2X2 subblock having the minimum of its diagonal elements greater than the absolute difference of its antidiagonal elements.”. It has offset 1 and begins

$$25396, 2814985, 304255924, 33102432698, 3594807908974, 390586327228567, 42432102644410392, 4609884118941$$

The entry states:

> Empirical recurrence of order 99 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 13:11:48, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 256$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 256$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 99: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 99.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 512$ exact terms from $n = 3$ on, it returns order 99 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{99} c_i \, a(n-i),$$

c_1	55	c_{26}	−4304381716065862571839312773 7	c_{51}	5958310419272
c_2	6936	c_{27}	−21624685631546653285903551269 5	c_{52}	23684097365498
c_3	−25361	c_{28}	942797952226221865889075201683	c_{53}	1651242076293
c_4	−10501395	c_{29}	2450853817887784678873230788448	c_{54}	−3424191400539
c_5	−49782183	c_{30}	−14965803807596483760737381464769	c_{55}	−6177451247161
c_6	7281846384	c_{31}	−19863890917723057321544596215145	c_{56}	40464228891745
c_7	47008656225	c_{32}	182317162574387199810190682646444	c_{57}	11579165500526
c_8	−2974113845417	c_{33}	101925221580817303609000442703247	c_{58}	−3864805305776
c_9	−15559242836013	c_{34}	−1760357670123822125999463003876748	c_{59}	−1482975180361
c_{10}	808182406393604	c_{35}	−1169370467790859807892670849373 86	c_{60}	29614043235942
c_{11}	1961553362673364	c_{36}	13767823408405280937181300125503940	c_{61}	13970945568512
c_{12}	−149291420783339817	c_{37}	−34159603813490571278871118308706 05	c_{62}	−1815203515841
c_{13}	129922552638988318	c_{38}	−886159743099098249017349514464801 01	c_{63}	−1000180282904
c_{14}	17827302704267460809	c_{39}	381176552132977770068155355129532 62	c_{64}	8901295209247
c_{15}	−70362315571600700040	c_{40}	4746058031128553846736245262590282 15	c_{65}	5543670173422
c_{16}	−1261116608269923028441	c_{41}	−2390657143111883041210624175707187 63	c_{66}	−3498754450943
c_{17}	854805615709751052780 3	c_{42}	−21268583948322774180792533122344805 00	c_{67}	−2408582585140
c_{18}	5148023194848157118340 7	c_{43}	103749696924518458383486972751019161 1	c_{68}	1104525771100
c_{19}	−557294070798954074037925	c_{44}	796871236404398833645928637191470927 2	c_{69}	8274659787111
c_{20}	−1011939857610264919855353	c_{45}	−3236221733751653434300812971610312663	c_{70}	−278865258284
c_{21}	2326829601428562105740426 0	c_{46}	−24811084699330137195022673798148238923	c_{71}	−225318154707
c_{22}	−86381915293305941459523 68	c_{47}	7161770306197856977858467237253220275	c_{72}	548345072805
c_{23}	−673089666664697917042902713	c_{48}	63772185404117491666749474938631937293	c_{73}	479176321060
c_{24}	1234628862492939734037024241	c_{49}	−104238013972010028799081543322624353 23	c_{74}	−76731138100
c_{25}	14058727134105339699905969551	c_{50}	−1351639010110393558211369309916255724 68	c_{75}	−75104900427

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 102$, and it is the recurrence OEIS A251227 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{99} - c_1 x^{99-1} - \cdots - c_{99}$. Its constant term is $-c_{99} = -30462644268000000$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 99 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 99, so $\deg q = 99$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 99, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 11 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 99, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 30462644268000000.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-99 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A251227.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] M. Kauers and P. Paule, *The Concrete Tetrahedron*, Springer, 2011. (C-finite sequences and their closure properties.)

[5] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.